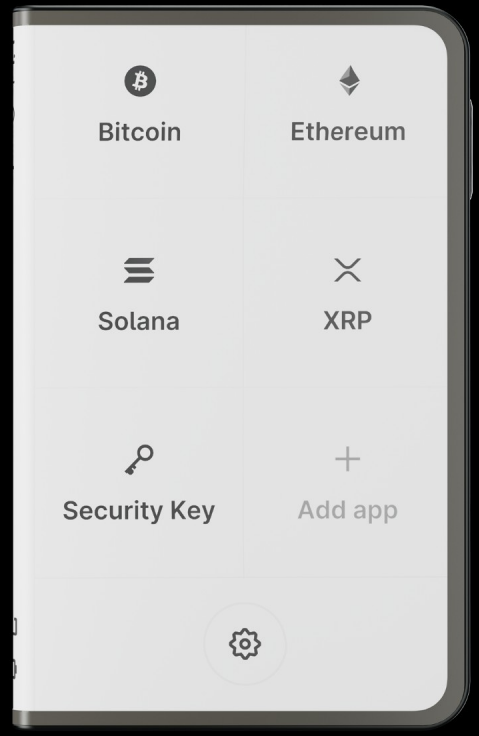
LEDGER

WORKSHOP · AI & FINTECH XRPL HACKATHON

Build on XRPL with Ledger

Sign XRPL transactions with the Device Management Kit, with or without a device



Olivier Freyssinet · Ledger · New York, October 24, 2026

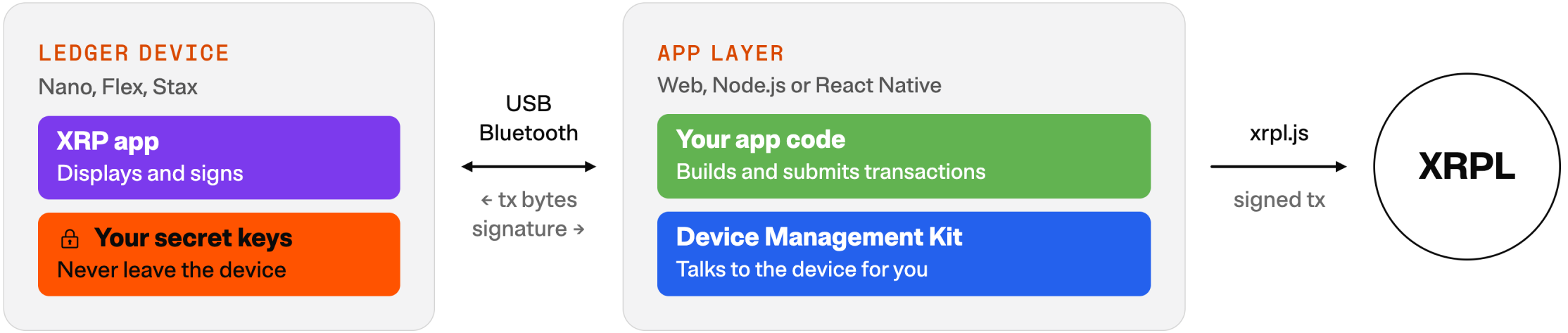
LEDGER

01 · BUILD ON TOP OF LEDGER

The tools you can use

The Device Management Kit, its XRP signer, and Speculos when you have no device

What lives where



Device Management Kit



- **Device discovery**
- **Connection**
- **Data exchange**
- **State management**
- **Transaction signing**
- **App management**

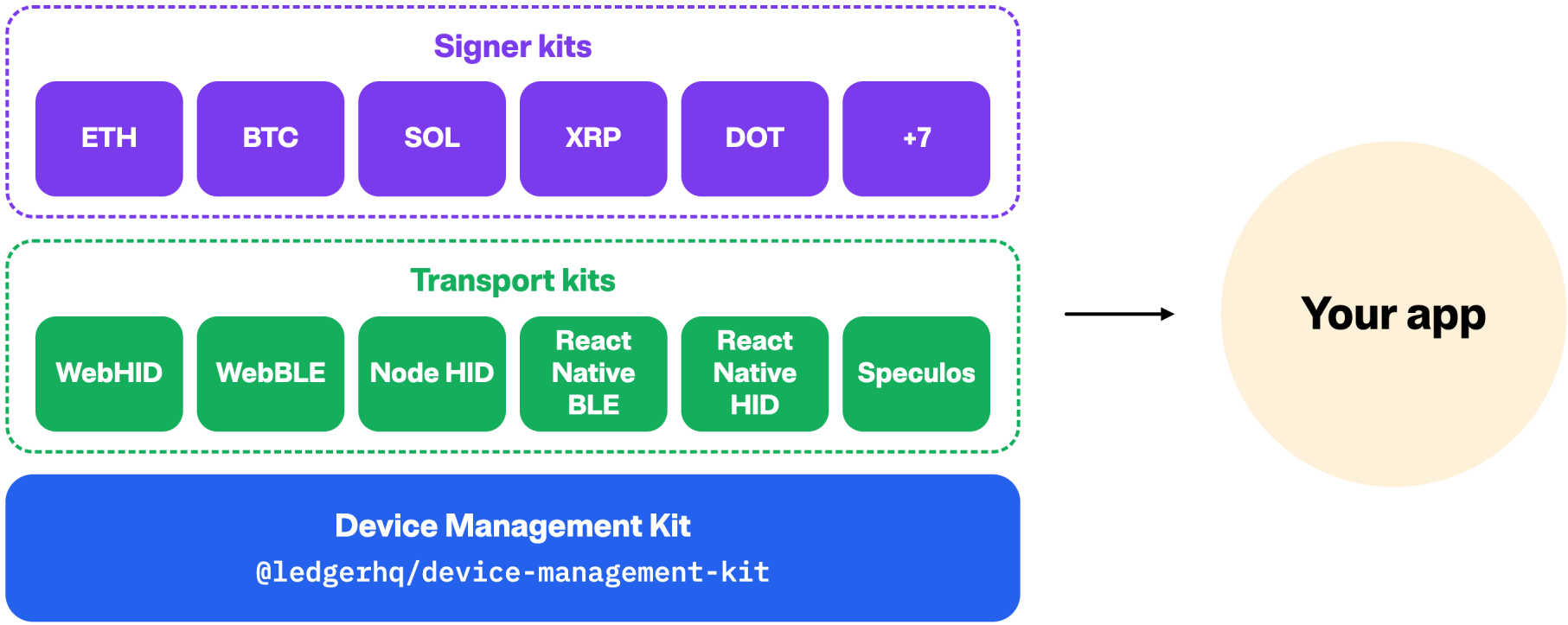
Try it app.devicesdk.ledger.com

npm `@ledgerhq/device-management-kit`

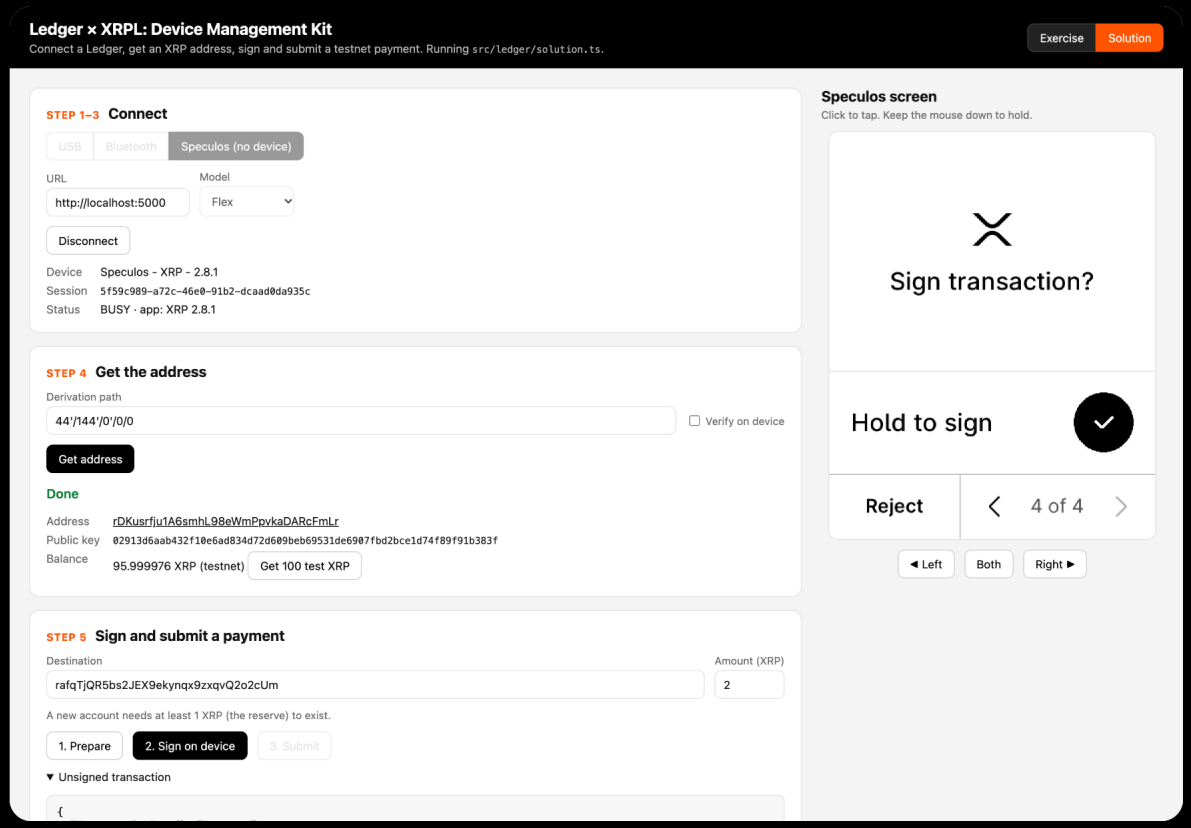
From LedgerJS to the DMK

	LedgerJS · hw-app-xrp LEGACY	DMK · XRP signer kit USE THIS
Abstraction	Low level: raw APDUs, status words	Device actions run the whole flow
Device state	Errors to catch, poll and retry	Live states: unlock, open app, sign
Opening the app	Up to you or the user	Done for you
Bluetooth	Legacy transports, unstable	Web BLE and React Native BLE
New features	Maintenance only, no new features	All new features, for XRP and others

A modular approach



Live demo: connect, sign, submit



1. Connect over USB, Bluetooth or Speculos
2. Get the address and 100 test XRP
3. Review and sign on the device
4. Submit to the XRPL testnet

github.com/LedgerHQ/device-management-kit-ts-workshop

Build the Device Management Kit 1/5

```
import { DeviceManagementKitBuilder } from "@ledgerhq/device-management-kit";
import { webHidTransportFactory } from "@ledgerhq/device-transport-kit-web-hid"; // + web-ble, speculos

const dmk = new DeviceManagementKitBuilder()
  .addTransport(webHidTransportFactory)        // USB
  .addTransport(webBleTransportFactory)        // Bluetooth
  .addTransport(speculosTransportFactory(url)) // Speculos emulator
  .build();
```

One instance for the whole app. On Node.js, add nodeHidTransportFactory instead.

Discover and connect 2/5

```
import { firstValueFrom } from "rxjs";
import { webHidIdentifier } from "@ledgerhq/device-transport-kit-web-hid";

const device = await firstValueFrom(
  dmk.startDiscovering({ transport: webHidIdentifier }),
);

const sessionId = await dmk.connect({ device });
```

Create the XRP signer 3/5

```
import { SignerXrpBuilder } from "@ledgerhq/device-signer-kit-xrp";

const signerXrp = new SignerXrpBuilder({ dmk, sessionId }).build();
```

Three methods: getAppConfig, getAddress, signTransaction. Each one opens the XRP app on the device first.

Get the XRP address 4/5

```
import { DeviceActionStatus } from "@ledgerhq/device-management-kit";

signerXrp.getAddress("44'/144'/0'/0/0", { checkOnDevice: true })
  .observable.subscribe((state) => {
    switch (state.status) {
      case DeviceActionStatus.Pending: // "unlock-device", "verify-address"…
        return showPrompt(state.intermediateValue.requiredUserInteraction);
      case DeviceActionStatus.Completed: // { address: "r…", publicKey }
        return setAccount(state.output);
      case DeviceActionStatus.Error:
        return showError(state.error);
    }
  });
```

Sign an XRPL transaction 5/5

```
import { DeviceActionStatus, hexaStringToBuffer } from "@ledgerhq/device-management-kit";

signerXrp.signTransaction("44'/144'/0'/0/0", hexaStringToBuffer(unsignedTx)!)
  .observable.subscribe((state) => {
    switch (state.status) {
      case DeviceActionStatus.Pending: // "sign-transaction": review on the device
        return showPrompt(state.intermediateValue.requiredUserInteraction);
      case DeviceActionStatus.Completed: // DER signature, Uint8Array
        return addSignatureAndSubmit(state.output);
      case DeviceActionStatus.Error:
        return showError(state.error);
    }
  });
```

Submit to the XRPL with xrpl.js

```
import { Client, encode, xrpToDrops } from "xrpl";
import { bufferToHexaString } from "@ledgerhq/device-management-kit";

const client = new Client("wss://s.altnet.rippletest.net:51233"); await client.connect();

const tx = await client.autofill({
  TransactionType: "Payment", Account: address, Destination: "r…", Amount: xrpToDrops("2"),
  SigningPubKey: publicKey.toUpperCase(), // the device's key, in uppercase
});
const unsignedTx = encode(tx);              // NOT encodeForSigning()
// … signerXrp.signTransaction(path, bytes) → signature
tx.TxnSignature = bufferToHexaString(signature, false).toUpperCase();
await client.submitAndWait(encode(tx));
```

What the XRP app signs today

SHOWN ON THE DEVICE

Clear signing

Payment (XRP and IOU tokens), TrustSet, Offers, Escrow, Checks, Payment channels, AccountSet, SignerListSet, Tickets, NFToken mint, burn, create and cancel offers, AMM, Clawback

OFF BY DEFAULT

Needs Blind signing

Credentials, Batch, DelegateSet, PermissionedDomains, DID, NFTokenModify, NFTokenAcceptOffer, AMMClawback

Also Payments and Offers that use DomainID, CredentialIDs or Delegate. Turn it on in the XRP app's settings.

REJECTED

Can't be signed

MPT tokens: payments in MPT and most MPT transactions. OracleSet. XChain bridges.

Also: at most 8 memos or signers per transaction, and no message signing.

Before you start

GOT A DEVICE

Ledger Wallet

Install the XRP app on your device, from Ledger Wallet.

ledger.com/ledger-wallet

NO DEVICE

Speculos

Run the real XRP app on your laptop, with Docker. Our workshop repo has it ready to run. Next slide.

`xrpl-hackathon/speculos`

BUILDING AN AGENT

Node.js

Agents, bots and backends: the same signer code, with the Node HID transport.

`device-transport-kit-node-hid`

Build on Ledger without a device

If you don't have a Ledger device, Speculos emulates a Ledger running the real XRP app, so you can develop without one.

```
$ docker pull ghcr.io/ledgerhq/speculos:latest
$ cd xrpl-hackathon/speculos
$ ./run-speculos.sh stax
```

```
import { DeviceManagementKitBuilder, DeviceModelId } from "@ledgerhq/device-management-kit";
import { speculosTransportFactory } from "@ledgerhq/device-transport-kit-speculos";

const dmk = new DeviceManagementKitBuilder()
  .addTransport(speculosTransportFactory("http://127.0.0.1:5000", false, DeviceModelId.STAX))
  .build();
```

⚠ Never use a real seed or real funds on Speculos. Its seed sits in plain text on your machine: use test seeds and testnet only.

LEDGER

02 · LEDGER BOUNTIES

$10,000

in Ledger bounties, across three tracks.

Pick your track

TRACK 01 · CORE

$4,000

Protocol Innovation

Bring the modern Ledger stack into XRPL, and lean into the gaps. A rough prototype that fills a real gap beats a polished rebuild.

- Put the new device-signer-kit-xrp to work: integrate it in a real app, find where it breaks, extend it. Bug reports and PRs count.
- Human-readable previews for TrustSet, OfferCreate, AMM or Escrow, before you sign.
- A Ledger-secured DEX, AMM or trust-line dApp for the flows Ledger Wallet can't do today.

TRACK 02

$3,000

Agentic Finance

Ledger as the trust layer for agents: device-backed security at the center.

- Agents that pay for APIs and services, with x402-style pay-per-use.
- Human in the loop: the Ledger approves high-risk actions before funds move.
- Agents that settle on XRPL: per-call micropayments or escrow-backed work, approved on the device.

TRACK 03

$3,000

Why Not

The open track: anything, as long as it settles at least one transaction on XRPL or the EVM sidechain.

- A polished consumer app or wallet experience that signs with a Ledger.
- Tooling that makes XRPL easier for people: address books, spending monitors, onboarding.
- Something fun: an on-chain game, a collectible, an interactive demo.

What we like

The same bar applies to all three tracks.

- Ledger used as a real trust layer, not just wallet branding.
- Signs on a Ledger device or on Speculos, against mainnet, testnet or the EVM sidechain.
- Fills a real gap rather than rebuilding something that already exists.
- A working demo we can run, not a slide deck.
- Clear boundaries between what is automated and what a human approves.
- Reusable by other developers where it makes sense: a library beats a one-off.

Documentation feedback

We judge the Developer Experience feedback as much as the code. Help us improve our docs.

What to include

- ✓ Feedback on overall experience using Ledger docs & SDKs
- ✓ Identify gaps, confusing flows, or missing context
- ✓ Specific improvements with screenshots or PRs

Bonus points

- \+ Better tutorial or code sample ideas
- \+ Navigation or search improvements for the portal
- \+ "Time-saver" suggestions for future integrators

LEDGER

FIND US ALL WEEKEND

Come build with us

Akram El Milligy	DevRel / Product Manager
Etienne Waldron	DevRel / Product Manager
Eva Mirza	Product Manager
Olivier Freyssinet	Software Engineer

Tracks, workshop code and slides

developers.ledger.com/**xrplnyc**